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CMMC Level 1 Media Protection (MP) Labs — Student Completion Guide

This guide provides step-by-step instructions for completing all 3 Media Protection labs. Each lab presents a real-world scenario involving removable media that contains Federal Contract Information (FCI): classifying it, sanitizing it for reuse, and deciding the correct disposition for end-of-life equipment.

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Before You Begin

What You Need

- Your **Pod number** (your instructor will assign this, e.g., Pod01, Pod05, Pod12)
- Your **Guacamole login credentials** (your instructor will provide your username and password)
- A computer with a web browser (Chrome, Firefox, or Edge) — no special software needed

What You Will Be Doing

In these labs you will work with **simulated removable media**. Each "USB drive" is a virtual disk file (`.vhdx`) that you mount on the domain controller exactly like a real thumb drive. You will inspect what is on the media, decide whether it contains FCI, sanitize media so it can be safely reused, and complete the paperwork (logs, certificates, worksheets) that an assessor would ask to see.

The key lesson of this family: deleting files is not sanitization. Your work is checked by inspecting the media itself, not just your paperwork.

CMMC Context

These labs align with **CMMC Level 1 Media Protection (MP)** requirements:

- **MP.L1-3.8.3** — Sanitize or destroy information system media containing Federal Contract Information before disposal or release for reuse

They also reinforce **ACS-POL-MP-001** (the company media handling policy referenced in your artifacts) and the media inventory/classification practices that support it.

How to Connect to the Lab Environment

You will connect to the lab through **Apache Guacamole** — a web-based remote desktop gateway. There is nothing to install; everything runs in your web browser.

Step 1: Open the Guacamole Gateway

1. Open your web browser (Chrome, Firefox, or Edge)
2. Go to: **https://crc.guac.01.tcecure.com/#/**
3. You will see a login screen

Step 2: Log In with Your Student Credentials

1. Enter your **Username** — this matches your pod number:
    - Pod 01 → `student01`
    - Pod 05 → `student05`
    - Pod 12 → `student12`
    - *(and so on — the number matches your assigned pod)*
2. Enter your **Password** (provided by your instructor)
3. Click **Login**

Step 3: Connect to the Domain Controller

Connection Name
PODXX-DC
PODXX-WS01

1. Click on **PODXX-DC** (where XX is your pod number, e.g., **POD03-DC**)
2. The remote desktop session opens directly in your browser — no extra login is needed
3. Wait a few seconds for the Windows Server desktop to appear

   Tip: Press **Ctrl+Alt+Shift** to open the Guacamole side menu (to return Home or switch connections).

Step 4: Check Your Lab Progress

Option 1: Click the "Check Your Progress — Pod XX" banner at the top of the Guacamole interface.

Option 2: Go directly to **https://training.status.tcecure.com/pod/XX** (replace XX with your pod number).

How to Open Your Lab Artifacts

1. Open **File Explorer** on the server desktop (folder icon on the taskbar, or **Windows + E**)
2. Navigate to: **C:\CyberLab\PodXX\MP-Artifacts** (e.g., `C:\CyberLab\Pod03\MP-Artifacts\`)

Alternative method using PowerShell:

1. Press **Windows key + R**, type `powershell`, press Enter
2. Run: `explorer "C:\CyberLab\Pod03\MP-Artifacts"` (replace `03` with your pod number)

What You Will See

File
MP-Lab-Instructions.txt
PXX-FCI-USB.vhdx
PXX-Employee-Handbook.vhdx
MediaInventory.xlsx
MediaClassificationWorksheet.docx
MediaClassificationResponses.csv
MediaDisposalPolicy.pdf
MediaSanitizationLog.csv
MediaSanitizationCertificate.csv

MP-M1-L2_SeedMetadata.json

LaptopAssetRecord.csv, ChainOfCustody.csv, VendorDestructionCertificate.txt

MP-M1-L3_DispositionWorksheet.csv

_LAB_READY_MP-M1-*.txt

> **Important:** Do not delete the original evidence files or the `_LAB_READY_` / `SeedMetadata` files. Fill in the CSV answer files in place (the checker reads those exact file names).

Editing the CSV Answer Files

The graded answer files are CSVs with the header row and blank cells already in place. Either:

- Open them with **Notepad** (right-click → *Open with* → *Notepad*) and type your values between the commas, or
- Open them with **Excel**, fill the cells, then **Save as CSV** (keep the same file name and `.csv` type)

Do not rename the columns, and do not add extra header rows.

Understanding Your Pod

Everything in your pod is prefixed with your pod number so your work stays separate from other students. If you are **Pod 03**, your prefix is **P03**:

- Artifacts folder: `C:\CyberLab\Pod03\MP-Artifacts\`
- FCI media: `P03-FCI-USB.vhdx` (media ID `P03-FCI-USB`)
- General media: `P03-Employee-Handbook.vhdx`
- End-of-life laptop: `P03-LAP-017`, drive serial `SN-P03-88421`
- Vendor destruction certificate: `VDC-P03-2026-017`

Throughout this guide, replace **XX** with your pod number and **PXX** with your pod prefix.

Working With the Lab Media (VHDX Files)

A `.vhdx` file behaves like a physical USB drive once you mount it.

Mount media (read it)

1. In File Explorer, open `C:\CyberLab\PodXX\MP-Artifacts\`
2. **Double-click** the `.vhdx` file (or right-click → **Mount**)
3. A new drive letter appears under **This PC** — open it to browse the contents

Show hidden files (you will need this in M1-L1)

In File Explorer: **View** ribbon → check **Hidden items**.

Or in PowerShell:

```
Get-ChildItem -Path E:\ -Recurse -Force
```

(Replace `E:\` with the drive letter that appeared when you mounted the media.)

Eject media

Right-click the drive under **This PC** → **Eject**.

Or in PowerShell:

```
Dismount-DiskImage -ImagePath "C:\CyberLab\PodXX\MP-Artifacts\PXX-FCI-USB.vhdx"
```

> **Tip:** Always eject media before moving on to the next lab so the checker can inspect it cleanly.

Module 1: Media Protection

Lab M1-L1: Classify the Media

Difficulty: Beginner | **Time:** 25 minutes | **Type:** Media inspection + worksheet

Scenario

Two USB drives were found in an unlocked desk drawer at ACS Consulting. Before either one can be reused, stored, or thrown away, you must determine whether it holds Federal Contract Information. FCI is information provided by or generated for the government under a contract that is **not** intended for public release.

What You Need

- `PXX-FCI-USB.vhdx`
- `PXX-Employee-Handbook.vhdx`
- `MediaInventory.xlsx`, `MediaClassificationWorksheet.docx`, `MediaDisposalPolicy.pdf`
- Answer file: `MediaClassificationResponses.csv`

Steps

1. **Read the policy first:**

   - Open `MediaDisposalPolicy.pdf` and note how the policy defines FCI and what handling it requires
2. **Mount and inspect the first drive:**
   - Double-click `PXX-FCI-USB.vhdx`
   - Turn on **Hidden items** in the View ribbon
   - Browse every folder: `Contracts`, `Purchase Orders`, `Drawings`, `Invoices`, `General Office`, and the hidden `Hidden Archive` and `Temp` folders
   - Open a few files. Look for the contract number `FA-2026-PXX` — that is a federal contract reference
   - Note that some FCI is hidden or in a temp file. Media is classified by the **most sensitive** thing on it, so one contract file is enough to make the whole drive FCI
3. **Mount and inspect the second drive:**
   - Double-click `PXX-Employee-Handbook.vhdx`
   - Review `Policies\Employee-Handbook.txt`, `Benefits\Benefits-Guide.txt`, `General Office\Holiday-Calendar.txt`
   - These are internal general business documents — no contract data, no government deliverables
4. **Complete the worksheet:**
   - Open `MediaClassificationWorksheet.docx` and record, for each drive, what you found and why it does or does not meet the FCI definition
   - Cross-check `MediaInventory.xlsx` so your media IDs match the inventory
5. **Record your graded answers in `MediaClassificationResponses.csv`:**

Column
Media
Classification
Evidence

   Example of a completed row (use your own findings):

   ```
   PXX-FCI-USB.vhdx,FCI,Contracts\Federal-Services-Contract-2026.txt and Invoices\INV-2026-031.txt reference federal contract FA-2026-PXX
   PXX-Employee-Handbook.vhdx,Non-FCI,Only general internal documents: employee handbook, benefits guide, holiday calendar
   ```

6. **Eject both drives** when finished

Completion Criteria

- ☐ `PXX-FCI-USB.vhdx` is classified as `FCI`
- ☐ `PXX-Employee-Handbook.vhdx` is classified as `Non-FCI`
- ☐ Both rows have specific written evidence (a short phrase is not enough)
- ☐ Worksheet, inventory, policy and both `.vhdx` files are still present in the folder

Why This Matters

You cannot protect FCI you have not identified. Media classification is the first step of MP.L1-3.8.3 — it decides which drives need sanitization or destruction later.

Lab M1-L2: Sanitize Media for Reuse

Difficulty: Intermediate | **Time:** 40 minutes | **Type:** Hands-on (disk management)

Scenario

`PXX-FCI-USB` is being reassigned to a non-federal project team. Before release for reuse it must be sanitized in accordance with **ACS-POL-MP-001**. A previous employee "sanitized" a drive by selecting the files and pressing Delete — the data was recovered by an auditor two weeks later. You will do it properly.

What You Need

- `PXX-FCI-USB.vhdx`
- Answer files: `MediaSanitizationLog.csv`, `MediaSanitizationCertificate.csv`
- Windows **Disk Management** (`diskmgmt.msc`) or PowerShell

Steps

1. **Understand what "sanitize" means:**
   - **Clear** — overwrite the media so data cannot be recovered with standard tools
   - **Purge** — a stronger method (e.g., degauss or cryptographic erase) that defeats laboratory recovery
   - **Destroy** — physically render the media unusable (used when media will not be reused)
   - Deleting files, emptying the Recycle Bin, and quick-deleting a folder are **none** of these
2. **Confirm what is on the media before you sanitize (evidence for your log):**

- Mount `PXX-FCI-USB.vhdx` and note the current volume label (`PXX-FCI-MEDIA`) and the folders present

3. **Eject the media, then re-create the volume.** The checker verifies that the volume itself was re-created — not merely emptied. Use one of the following.

   Option A — Disk Management (GUI):

   - Press **Windows + R**, type `diskmgmt.msc`, press Enter
   - **Action → Attach VHD**, browse to `C:\CyberLab\PodXX\MP-Artifacts\PXX-FCI-USB.vhdx`, click OK
   - Find the attached disk at the bottom of the window (it will be about 96 MB)
   - Right-click its partition → **Delete Volume** → Yes
   - Right-click the resulting unallocated space → **New Simple Volume** → Next
   - Use the full size → assign any drive letter → **Format this volume**:
     - File system: **NTFS**
     - Volume label: **PXX-SANITIZED** (exactly this, with your pod prefix)
     - **Uncheck** "Perform a quick format" (a full format overwrites the data area)
   - Finish and wait for the format to complete
   - Right-click the disk (left-hand grey box) → **Detach VHD**

   Option B — PowerShell:

```
$vhd = "C:\CyberLab\PodXX\MP-Artifacts\PXX-FCI-USB.vhdx"

Mount-DiskImage -ImagePath $vhd
$disk = Get-DiskImage -ImagePath $vhd | Get-Disk

# Remove the old volume and create a brand-new one
Clear-Disk -Number $disk.Number -RemoveData -Confirm:$false
Initialize-Disk -Number $disk.Number -PartitionStyle MBR -
ErrorAction SilentlyContinue
$part = New-Partition -DiskNumber $disk.Number -UseMaximumSize -
AssignDriveLetter
Format-Volume -Partition $part -FileSystem NTFS -NewFileSystemLabel
"PXX-SANITIZED" -Full -Confirm:$false

# Confirm the volume is empty
Get-ChildItem -Path "$($part.DriveLetter):\" -Force

Dismount-DiskImage -ImagePath $vhd
```

(Replace `PodXX` / `PXX` with your pod values. `-Full` performs the overwriting format.)

4. **Validate your work before writing the paperwork:**
   - Re-mount the media and confirm the label reads `PXX-SANITIZED`
   - Confirm the volume is empty — `System Volume Information` and `$RECYCLE.BIN` are the only things allowed
   - Eject the media
5. **Complete `MediaSanitizationLog.csv` and `MediaSanitizationCertificate.csv`.** Both files use the same columns and both must be filled in:

Column
MediaId
Method
Result
Disposition
SanitizedBy
Date

Completion Criteria

- ☐ The volume was **deleted and re-created** (not just emptied)
- ☐ Volume label is exactly `PXX-SANITIZED`
- ☐ The volume contains no files or folders
- ☐ Sanitization **log** and **certificate** both record `Clear` or `Purge`, `Pass`, `Reuse`, a `SanitizedBy` name, and a valid date
- ☐ `MP-M1-L2_SeedMetadata.json` is still present and unmodified

  Hint: If the checker says *"volume identity is unchanged; recreate and format the volume rather than deleting files"*, you emptied the drive instead of re-creating the volume. Repeat step 3.

Why This Matters

MP.L1-3.8.3 requires sanitization **before** media is released for reuse. Deleted files remain recoverable, so an assessor tests the media and the records — which is exactly what this lab does.

Lab M1-L3: Decide the Disposition

Difficulty: Intermediate | **Time:** 25 minutes | **Type:** Document analysis + worksheet

Scenario

Laptop `PXX-LAP-017` from the Federal Programs group has failed diagnostics and reached end of life. Its drive held FCI. Facilities has already sent it out with a chain-of-custody record, and the vendor returned a certificate. Your job is to record the correct disposition decision and prove that the paperwork lines up.

What You Need

- `LaptopAssetRecord.csv`
- `ChainOfCustody.csv`
- `VendorDestructionCertificate.txt`
- Answer file: `MP-M1-L3_DispositionWorksheet.csv`

Steps

1. **Read the asset record** (`LaptopAssetRecord.csv`):
   - Asset `PXX-LAP-017`, owner Federal Programs, status **End of life**
   - `ContainsFCI` = **Yes**
   - `DriveSerial` = `SN-PXX-88421`
   - `Serviceability` = **Failed diagnostics**
2. **Read the chain of custody** (`ChainOfCustody.csv`):
   - Record `COC-PXX-017` shows the drive released by Jordan Lee to an approved destruction vendor for the purpose of **Destruction**
   - Confirm the drive serial matches the asset record
3. **Read the vendor certificate** (`VendorDestructionCertificate.txt`):
   - Certificate `VDC-PXX-2026-017`, method **Physical shredding**, status **DESTROYED**, witnessed
   - Confirm the asset ID and drive serial match
4. **Reason through the decision:**
   - **Reuse** is wrong — the drive failed diagnostics and cannot be reliably sanitized or trusted
   - **Return to inventory** is wrong — the asset is end of life and still contains FCI
   - **Destroy** is correct — FCI media that will not be reused must be destroyed, and the custody record and vendor certificate already document destruction
5. **Complete `MP-M1-L3_DispositionWorksheet.csv`:**

Column
AssetId
Decision

DriveSerial

CertificateId

Rationale

Example rationale:

```
Drive failed diagnostics and cannot be reliably sanitized; the
asset is end of life and held FCI, so destruction was performed and
documented on certificate VDC-PXX-2026-017.
```

Completion Criteria

- ☐ `Decision` is `Destroy`
- ☐ `DriveSerial` and `CertificateId` match the supporting records exactly
- ☐ `Rationale` is a complete justification, not a single word
- ☐ Asset record, chain of custody and vendor certificate are all still present

Why This Matters

MP.L1-3.8.3 gives you two valid outcomes for FCI media: sanitize it or destroy it. Choosing correctly — and being able to show matching custody and certificate records — is what turns a decision into audit evidence.

Quick Reference: Where to Find Things

Task
Open the artifacts folder
Mount simulated media
Eject simulated media
Show hidden files
Delete/create/format a volume
Open PowerShell
Check volume label and contents
Check your lab progress

Media Handling Terms

Term
FCI
Clear
Purge
Destroy
Chain of custody

Tips and Common Mistakes

Mistake
Deleting files and calling it sanitization
Quick format instead of full format
Wrong volume label
Filling in only the sanitization log
Leaving Result as Pending
One-word evidence or rationale
Media left mounted
Editing or deleting MP-M1-L2_SeedMetadata.json
Renaming the answer CSVs
Missing hidden FCI in M1-L1

Getting Unstuck

- **Cannot mount the VHDX?** It may already be attached — check **This PC**, or run `Dismount-DiskImage` and try again
- **Verification still failing?** Re-mount the media and check the label and that the volume is empty; then re-read the exact wording of the failure reason on your progress page
- **Broke the media?** Ask your instructor to reseed the MP family for your pod

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Lab Completion Checklist

Lab
M1-L1
M1-L2
M1-L3

After completing each lab:

1. Confirm your answer CSVs are saved in `C:\CyberLab\PodXX\MP-Artifacts\` with their original names
2. Confirm all seeded evidence files are still present
3. Eject any mounted media
4. Check your status at https://training.status.tcecure.com/pod/XX

This guide was created for the Digital Resilience Community Clinic (DRCC) Cyber Range. CMMC Level 1 — Media Protection (MP) Module